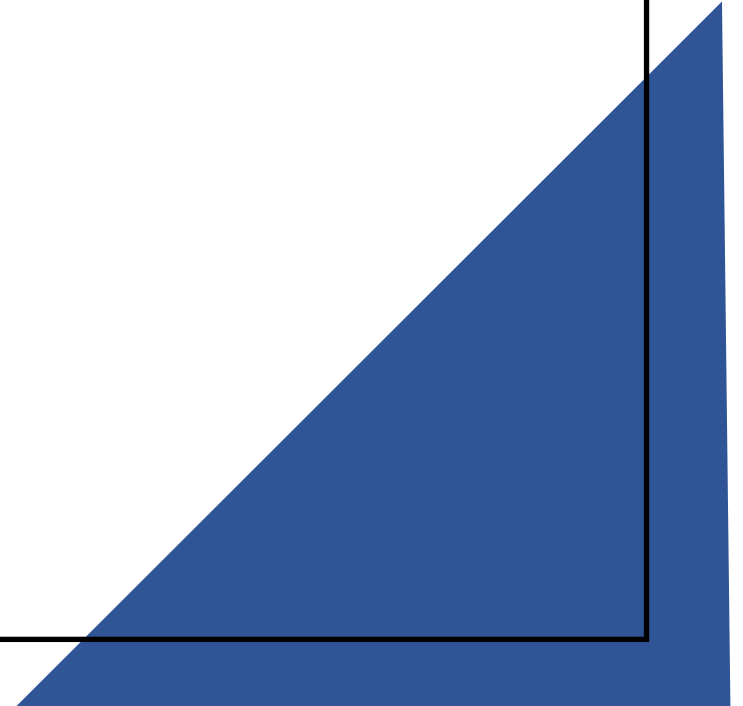


Mathematical Foundations

CSCE 411

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Questions?

- Any questions on the reading?
- Likely some of them will be answered through the problems I've prepared, but knowing what was confusing helps me know what to focus on
- And you may have questions I didn't anticipate!

Learning Objectives

- Become comfortable manipulating:
 - Sets
 - Exponents and Logarithms
 - Summations and Products
 - Floors and Ceilings

AI Use

- AI is not necessary to learn this material
- Nevertheless, you may find it useful
- AI agents can provide personalized help/tutoring on-demand
- But it's also easy to over-rely on them in a way that hinders your learning
 - Remember our discussion of active learning: you understand and remember something much better when you figure it out yourself instead of just passively having it explained to you
 - Some amount of struggle is important for learning
- You can access a variety of LLMs for free through TAMU AI Chat (chat.tamu.ai)
 - Doesn't include the most cutting edge models like GPT 5.6 sol or Claude Fable
 - But the models available are pretty competent at undergraduate algorithms

Helpful AI Prompts¹

- It may be useful to include context like “I am an undergraduate CS student”
- “What are some good sources to learn about X?”
 - LLMs tools can be effective as enhanced search engines, pointing you towards sources to learn about a particular topic that will be more reliable than the AI itself
- “I have been reading X (upload a file) and on page Y I have trouble understanding the proof of Theorem Z. Could you explain the proof in more detail?”
- “I am trying to solve this problem [copy or upload a file] and I have attached my attempted solution. Could you please proofread it for me?”
 - I’ve found this to work well even with scans of handwritten proofs

¹ Some of these prompts are taken from Pavel Etingof at MIT

Unhelpful AI Prompts

- “Summarize this chapter for me”
 - Summarizing papers with AI can be helpful if all you care about is the results
 - Not so good if you need to learn the techniques
 - Identifying the core ideas in a chapter and connecting them to each other is an important part of learning
- “Solve this exercise”
 - By spoiling yourself on the answer you’re missing out on the chance to practice solving the problem yourself
 - “Solving” the exercises this way to prepare for exams is like training for a marathon on a motorcycle
 - AI solutions may also use results that you can’t assume (i.e. true theorems that we haven’t covered)

AI Mistakes

- Here's an actual quote from Claude Opus 4.8 when I asked it for feedback on Ch1 of the course notes this summer:

- “The notes state:

$$'(c) \log_b x = (\log_c x) \log_b c'$$

This is wrong as written. The standard change of base identity is

$$\log_b x = \frac{\log_c x}{\log_c b}$$

The version in the notes would only be correct if it were $\log_b x = (\log_c x)(\log_c b)^{-1}$, i.e., a *division*, not a product.”

Why is Claude completely wrong?

Manipulating the Change of Base Formula

My version in the notes:

$$\log_b x = (\log_c x) \log_b c$$

Claude's version:

$$\log_b x = \frac{\log_c x}{\log_c b}$$

- If we divide both sides of the formula in the notes by $\log_b c$, we get $\frac{\log_b x}{\log_b c} = \log_c x$
- Swapping b and c (they were both defined as arbitrary positive real number not equal to one, so it doesn't matter which is which), we get $\frac{\log_c x}{\log_c b} = \log_b x$, which is Claude's formula.
- The two forms are exactly equivalent, and saying we need to use a division is nonsense
- It's important to verify AI output yourself, because LLMs will confidently lie to you

AI “Hallucination”

- AI wants to tell you what you want to hear
- If it doesn't know the right answer, it'll often make its best guess and present it as if it's completely confident (kind of like students on an exam ^_-)
- If you ask a recent LLM to proofread a solution with a major problem, it will very likely find the problem
- If you ask it to proofread a completely correct solution, it may still make up a problem
 - More training examples of proofreaders finding problems than of saying “yes, this is completely fine”
- If you aren't sure whether the feedback you've gotten is legit, please ask me or Ryan

Sets

- Ways to describe a set:
 - List the element directly, e.g. $\{2, 3, 4, 5\}$
 - Set Builder notation:
 - $\{x \in S \mid P(x)\}$: “all x in S for which $P(x)$ is true”
 - $\{f(x) \mid x \in S\}$: “the set containing $f(x)$ for each x in S ”
 - Note that the universal set S is essential here, $\{x \mid P(x)\}$ is not a valid set
 - Otherwise we could create paradoxical sets like $\{x \mid x \notin x\}$ (Russell’s paradox)
 - Combination of existing sets, e.g. $\mathbb{Z}^+ \cup \{-2, -3\}, \mathbb{Z} \times \mathbb{Z}$

Building Sets

- What is $\left\{ x \in \mathbb{Z} \mid \frac{x}{3} \in \mathbb{Z} \right\}$?
 - The set of multiples of 3
- What is $\{ 3x \mid x \in \mathbb{Z} \}$?
 - Also the set of multiples of 3!
- The same set can be defined in many different ways
- What is $\{ 3x \mid x \in \{1,2\} \}$?
 - $\{3, 6\}$
- What is $\left\{ x \in \{1,2\} \mid \frac{x}{3} \in \mathbb{Z} \right\}$?
 - $\emptyset$
- Pay attention to the set being quantified over, and watch out for differences between similar-looking sets

Powersets

What is $|2^S|$ in terms of $|S|$?

- Answer: $2^{|S|}$
- Why is this always true?
- For each element of S , we have two choices: include it or skip it
- Each possible sequence of choices gives a different subset
- e.g. consider $\{2,5\} \in 2^{\{1,2,3,4,5\}}$
 - This set corresponds to making the choices: skip, include, skip, skip, include

Exponents and Logarithms

- Formulas involving exponents and logarithms arise often in analyzing algorithms
- It's important to be able to manipulate and compare these expressions

Comparing Exponentials

Which grows faster, 4^n or 2^{2n} ?

- A good way to get intuition for what the right answer should be is to try small examples
 - If $n = 1$, $4^n = 4$ and $2^{2n} = 4$
 - If $n = 2$, $4^n = 16$ and $2^{2n} = 16$
 - If $n = 3$, $4^n = 64$ and $2^{2n} = 64$
- It seems like these two are actually the same function!
- How can we confirm this?
- Since $4 = 2^2$, $4^n = (2^2)^n = 2^{2n}$
- Rewriting exponents to have a common base (2 in this case) helps us compare them

Simplifying Exponentials

Which grows faster, n^3 or $(2n)^2$?

- At $n = 1$, $1 < 4$
- At $n = 2$, $8 < 16$
- At $n = 3$, $27 < 36$
- Should we conclude that n^3 is always smaller?
- No!
- In experimental sciences, sometimes “it worked for a bunch of examples” is the best evidence we can get that something is true
- But in CS theory (like in the rest of math), examples are not enough, we need a proof
- Lots of theorems that hold for small numbers fail for big ones

Simplifying Exponentials

Which grows faster, n^3 or $(2n)^2$?

- How do we relate these quantities?
- The order in which we multiply numbers doesn't change the result (multiplication is commutative and associative)
- We can rewrite $(2n)^2 = (2n)(2n)$ as $2 \cdot 2 \cdot n \cdot n = 2^2 n^2 = 4n^2$
- Meanwhile $n^3 = n \cdot n^2$
- So $n^3 > (2n)^2$ if and only if $n > 4$

Logarithms

Definition: $\log_b a$ $\coloneqq$ the power that we have to raise b to in order to get a

- By definition, $b^{\log_b a} = a$
- Exponents and logarithms “cancel each other out”
- We have to pay attention to the base b , however
- Sometimes, particularly when analysing divide and conquer algorithms, we’ll get expressions like $3^{\log_5 n}$
- How do we simplify this?
- In particular, how can we write it in the form n^c , for some constant c
- Expressing it in this form would make it easy to see compare this function with familiar polynomials like n or n^2

Simplifying Expressions with Logarithms

Simplify $3^{\log_5 n}$

Theorem 2(c): $\log_b x = (\log_c x) \log_b c$

Theorem 1(b): $(x^a)^b = x^{ab}$

- We want to get the exponential and the logarithm to cancel, so we want to convert our base 5 log into a base 3 log
- By our theorem, $\log_5 n = (\log_3 n) \log_5 3$
- Thus, $3^{\log_5 n} = 3^{(\log_3 n) \log_5 3}$
- $3^{(\log_3 n) \log_5 3} = (3^{\log_3 n})^{\log_5 3} = n^{\log_5 3} \approx n^{0.683}$

Summations

Theorem 3: $\sum_{i=1}^n i = \frac{n(n+1)}{2}$

Give a formula for $\sum_{i=10}^n i$.

- Good to start by considering small examples:
 - If $n < 10$, the sum is 0 (the $i > 10$ stopping condition is satisfied immediately, and we define the sum of nothing as 0)
 - If $n = 10$, the sum is 10
 - If $n = 11$, the sum is $10+11 = 21$
 - If $n = 12$, the sum is $10+11+12 = 33$
 - We can use these examples to check that whatever formula we derive is correct

Finding a Formula

Theorem 3: $\sum_{i=1}^n i = \frac{n(n+1)}{2}$

Give a formula for $\sum_{i=10}^n i$.

- How can we relate this new sum to the sum we know?
- $\sum_{i=1}^n i = (\sum_{i=1}^9 i) + (\sum_{i=10}^n i)$ [assuming $i \geq 10$]
- Applying Theorem 3, we have: $\frac{n(n+1)}{2} = \frac{9(10)}{2} + \sum_{i=10}^n i$
- Subtracting $\frac{9(10)}{2} = 45$ from both sides we have:
- $\sum_{i=10}^n i = \frac{n(n+1)}{2} - 45$ if $i \geq 10$, or 0 if $i < 10$
- Checking $n = 10, 11, 12$ against the values we computed before, we can confirm that this formula works

Another Summation

Theorem 3: $\sum_{i=1}^n i = \frac{n(n+1)}{2}$

Give a formula for $\sum_{i=1}^n 2i + 3$.

- Good to start by considering small examples:
 - If $n = 1$, the sum is 5
 - If $n = 2$, the sum is $5+7=12$
 - If $n = 3$, the sum is $5+7+9 = 21$
- Since changing the order in which we add numbers doesn't change the result, we can split this sum into two:
- $\sum_{i=1}^n 2i + 3 = (\sum_{i=1}^n 2i) + (\sum_{i=1}^n 3)$

Simplifying the Summation

Theorem 3: $\sum_{i=1}^n i = \frac{n(n+1)}{2}$

Give a formula for $\sum_{i=1}^n 2i + 3$.

- $\sum_{i=1}^n 2i + 3 = (\sum_{i=1}^n 2i) + (\sum_{i=1}^n 3)$
- What is $\sum_{i=1}^n 2i$?
 - Since each term is double the corresponding term in $\sum_{i=1}^n i$, the overall sum will also be twice as large
 - i.e. $\sum_{i=1}^n 2i = 2 \sum_{i=1}^n i = n(n+1)$
 - The fancy way to say this is “multiplication distributes over addition”
- What is $\sum_{i=1}^n 3$?
 - We’re summing n 3s, so it’s just $3n$
- Thus, $\sum_{i=1}^n 2i + 3 = n(n+1) + 3n = n(n+4)$

Re-indexing

Theorem 3: $\sum_{i=1}^n i = \frac{n(n+1)}{2}$

Give a formula for $\sum_{i=10}^n i - 9$.

- We could combine the techniques on the previous slides:
- $\sum_{i=10}^n i - 9 = (\sum_{i=10}^n i) - \sum_{i=10}^n 9 = (\sum_{i=1}^n i) - (\sum_{i=1}^9 i) - 9(n - 9)$
- $= \frac{n(n+1)}{2} - 45 - 9n + 81 = \frac{n(n-17)}{2} + 36$
- Alternatively, we could *re-index* the sum by defining $j := i - 9$
- Then the sum becomes $\sum_{j=1}^{n-9} j$
- So we can directly apply Theorem 3 to get the formula $\frac{(n-9)(n-8)}{2}$
 - This is equivalent to the formula we found using our previous techniques

Floors and Ceilings

- The floor and ceiling functions are useful when we need to upper or lower bound a potentially non-integer quantity with integers
- For every real number x , we have $x - 1 < \lfloor x \rfloor \leq x \leq \lceil x \rceil < x + 1$

Prove that for every positive integer $n \geq 4$, $\left\lfloor \frac{n}{2} \right\rfloor > \frac{n}{3}$.

- From the observation above, we have that $\frac{n}{2} - 1 < \left\lfloor \frac{n}{2} \right\rfloor$, but this isn't quite good enough to say what we want (e.g. $4/2 - 1$ is $< 4/3$)
- Instead, we note that for $n \geq 4$, $\left\lfloor \frac{n}{2} \right\rfloor \geq \frac{n}{2} - \frac{1}{2} \geq \frac{n}{2} - \frac{n}{8} = \frac{3}{8}n > \frac{n}{3}$

Modular Arithmetic

- Floors are also helpful when working with modular arithmetic
- In fact, for an integer a and positive integer n , $a \bmod n := a - n \left\lfloor \frac{a}{n} \right\rfloor$
- We'll see an application of this next week

Summary

- Today:
 - Sets
 - Exponents
 - Logarithms
 - Summations
 - Floors and Ceilings
- Next time:
 - What is a proof?
 - How do we write one?